

Sonali Godavarthy

AI Researcher

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📍 Siegen, Germany (open to relocate) 🔗 LinkedIn 🐙 GitHub 🏠 Google Scholar



PROFILE

AI Researcher focused on Generative AI, Computer Vision, and Foundation Models. Experienced in developing novel methods, benchmarks, and evaluation frameworks for controllable image generation and multimodal activity recognition, with hands-on experience in large-scale experimentation using PyTorch, and SLURM/HPC.

EDUCATION

B.E. Information Technology, Vasavi College of Engineering 07/2018 – 08/2022 | Hyderabad, India
Thesis: Voice Control and Command System for Unmanned Aerial and Ground Vehicles
GPA: 1.5

M.Sc. Computer Science (Visual Computing), University of Siegen 10/2023 – 08/2026 | Siegen, Germany
Thesis: Incremental Learning for Disentangling of Multiple Visual Concepts for Text-to-Image
GPA: 1.6 (1.1 Thesis Grade)

PUBLICATIONS

- S. Godavarthy et al., “MULTI: Disentangling Camera Lens, Sensor, View, and Domain for Novel Image Generation,” *International Conference on Pattern Recognition (ICPR)*, 2026. **Accepted for Oral Presentation.** [🔗](#)
- S. Godavarthy et al., “X-MULTI: VLM-based Imaging Factor Disentanglement for Factor-Aware Image Synthesis,” *2nd Workshop on Multimodal Large Language Models for Unified Comprehension and Generation (MUCG), European Conference on Computer Vision (ECCV)*, 2026. **Accepted Paper.**

RESEARCH EXPERIENCE

University of Siegen, AI Researcher 04/2026 – Present | Siegen, Germany

- Collaborating within an interdisciplinary research team to advance multimodal Human Activity Recognition (HAR) by exploring inertial measurement units (IMUs) and vision-based foundational models (CLIP, DINOv2, DINOv3).
- Engineered an automated annotation pipeline that leverages feature extraction from vision and IMU foundational models to label sensor streams, achieving a 10% improvement in annotation accuracy over baseline methods.
- Conducting downstream activity recognition and classification studies to evaluate the impact of the refined annotations on final model generalization and accuracy.

Robert Bosch GmbH, AI Researcher 09/2025 – 03/2026 | Hildesheim, Germany

- Collaborated with an interdisciplinary research team from Bosch Research and ETH Zurich to conduct master thesis on text-to-image, multimodal foundational methods and continual learning.
- Proposed and implemented a novel multi-embedding architecture aimed at disentangling visual concepts; achieved a ~10% improvement in factor disentanglement compared to established baseline models.
- Designed and established a standardised benchmark to quantitatively evaluate the adaptability of generative systems to unseen visual concepts.
- Executed large-scale model training and ablation studies using HPC environments (SLURM/multi-GPU) to validate the scalability and robustness of the proposed method.

Robert Bosch GmbH, AI Researcher 04/2025 – 09/2025 | Hildesheim, Germany

- Conducted a technical investigation into latent space disentanglement within foundational models to identify limitations in current multimodal synthesis as an intern.
- Developed an embedding-based multimodal prototype that optimized model efficiency and increased disentanglement accuracy by ~5%.
- Evaluated the proposed method across various downstream generative tasks, focusing on the trade-offs between computational efficiency and image fidelity.

Fraunhofer Institute for Mechatronic Systems Design 06/2024 – 03/2025 | Paderborn, Germany
AI Research Engineer (GenAI Incubator)

- Conducted research on OCR tasks and developed an Information Extraction system, leveraging the Flamingo OCR and GPT-4o API.
- Implemented few-shot prompting and system-level optimizations that reduced misclassification rates by ~15% and significantly decreased manual review overhead by ~70%.

- Built a frontend using ReactJS to facilitate the visualization, annotation, and benchmarking of OCR results for experimental analysis.

Advanced Data Processing Research Institute (ADRIN), *Research Intern*

01/2022 – 06/2022 | Hyderabad, India

- Researched Voice Command control and developed a Voice Control and Command System for Unmanned Aerial and Ground Vehicles (UAV/UGV).
- Built a AttRNN model using TensorFlow, achieving 95% accuracy on the Google Speech Commands dataset.
- Implemented real-time remote control by deploying the model on a Raspberry Pi and establishing secure communication via ZeroTier VPN.

PROFESSIONAL EXPERIENCE

S&P Capital IQ Private Limited, *Engineer 1, Site Reliability Engineer*

08/2022 – 09/2023 | Hyderabad, India

- Built and scaled AWS infrastructure with Terraform across 10+ services, sustaining 99.9%+ uptime.
- Engineered Datadog-based observability, cutting Mean Time to Detection by 70%.
- Automated cost, metric, and incident-reporting pipelines within a 7-member global infrastructure team.

Brane Services Private Limited, *Project Intern*

05//2022 – 07//2022 | Hyderabad, India

- Designed 20+ validation test cases and QA checklists, raising application test coverage by 80% across three release cycles.
- Tracked and validated 30+ application defects with a 5-member team, shortening release verification effort by 30%.

SKILLS

Research Areas Generative AI, Diffusion Models, Computer Vision, Multimodal Learning, Foundation Models, Human Activity Recognition, Controllable Image Generation	Generative AI & Diffusion Models Stable Diffusion, FLUX, Diffusers, LoRA, QLoRA, BLIP	Computer Vision & Vision-Language Models OpenCV, CLIP, DINOv2, DINOv3, SAM
Machine Learning & Deep Learning PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn	Mathematical Foundations Linear Algebra, Probability & Statistics, Calculus, Optimization	Programming & Data Python, SQL, NumPy, Pandas, Matplotlib, Seaborn
Cloud & Databases AWS, Oracle SQL, MySQL	Experimentation & Infrastructure Git, Conda, LaTeX, Docker, Kubernetes, MLflow, SLURM	Development Tools Git, GitHub, Claude Code, Conda, LaTeX

AWARDS AND ACHIEVEMENTS

Deutschlandstipendium, *Studienförderfonds Siegen e.V.*

10/2024

Merit-based scholarship supporting academic excellence and commitment awarded to top ~1% of students in Germany

Best Project of 2021-22, *Vasavi College of Engineering*

07/2022

1st Prize for Best Final Year Project (~120 competing teams)

RESEARCH PROJECTS

Jigsaw Puzzle Solver using Computer Vision

- Developed an end-to-end computer vision pipeline utilizing QR-based projective transforms for coordinate rectification and Segment Anything Model (SAM) for zero-shot segmentation of individual puzzle pieces.
- Implemented automated rotation correction and utilized specialized OpenCV algorithms for piece identification, including shape-based corner detection and color-profile matching.
- Engineered a graph-based matching algorithm combined with recursive backtracking to perform robust piece alignment and solve ~80% of complex, large-scale puzzles.

Denoise Microscopic Data with Deep Learning

- Investigated deep learning-based denoising techniques for microscopic imaging by implementing a UNet architecture inspired by Blind Spot Denoising.
- Conducted a comparative analysis between deep learning approaches and classical filters (Gaussian, Median, Box3D), quantifying a ~54% PSNR improvement in image fidelity.
- Validated and benchmarked model performance on specialized biological datasets, including LiveCell and Mouse Actin.

Christmas Image Classification

- Performed an empirical study on transfer learning by fine-tuning GoogLeNet in PyTorch for high-accuracy domain-specific image classification.
- Executed an architectural benchmarking study comparing AlexNet, ResNet, and GoogLeNet, optimizing model selection to achieve a 91% classification accuracy.
- Quantified the impact of hyperparameter tuning and curated datasets, resulting in an 8–12% accuracy gain over baseline architectural configurations.

German Legal AI Assistant & Enterprise RAG Engine

- Assembled a multilingual (German/English) RAG pipeline over 9,000+ metadata-tagged legal document chunks using MiniLM-L12 embeddings and citation-grounded response generation.
- Implemented semantic-hallucination monitoring with Langfuse and 40 automated tests, achieving sub-200ms latency on metadata-aware search.

AskDoc - Research Paper RAG Assistant

- Developed a document QA system converting research papers into searchable embeddings for natural-language querying, with source-grounded answer generation over FAISS vector search.

LANGUAGES

English | **German** — Conversational (B1) | **Hindi**